

PRODUCTION GATE

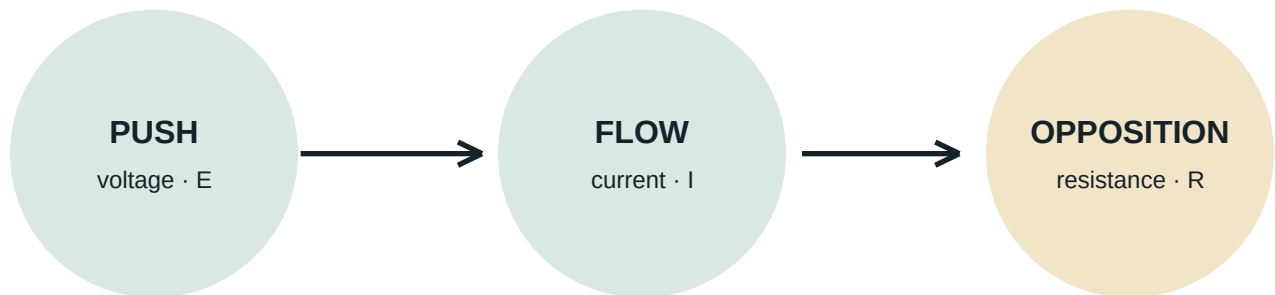
OHM'S LAW WITHOUT THE FOG

A 14-page reading-experience prototype for the Ham Radio
Technician Visual Cram Map 2026–2030

One relationship. Three quantities. Sixteen official question IDs made easier to recognize.

The map before the math

A circuit question usually gives you two quantities and asks for the third. Do not begin with a triangle or a calculator. Begin with the story.



The single mental model

More electrical push produces more flow. More opposition produces less flow. Ohm's law turns that relationship into arithmetic.

POOL LINKS

T5A05 · T5A11 · T5D01–T5D12

Name the three quantities

Voltage (E), measured in volts

The electrical potential difference—the push available between two points.

Current (I), measured in amperes

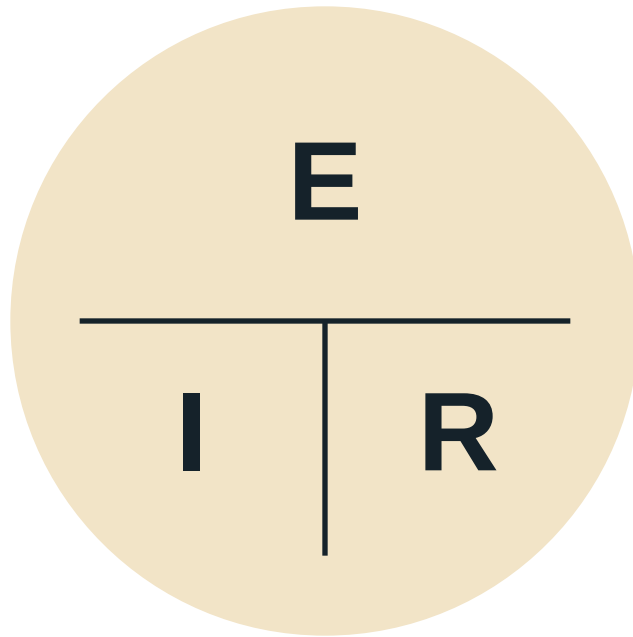
The rate of charge flow through a path. Current is what moves through a component.

Resistance (R), measured in ohms

How strongly a component opposes current. More resistance means less current when voltage stays fixed.

Exam-reading move: circle the two units you are given. The missing unit tells you which form of Ohm's law to use.

Choose the formula from the missing piece



Need current?

$$I = E \div R$$

Need voltage?

$$E = I \times R$$

Need resistance?

$$R = E \div I$$

Memory aid, not magic: the diagram works because the three formulas describe the same relationship.

Worked pattern: find resistance

1 · FIND THE GIVEN

Given: 12 volts and 4 amperes

2 · NAME THE MISSING UNIT

Missing: ohms → resistance

3 · CHOOSE AND SOLVE

$$R = E \div I$$

$$R = 12 \text{ V} \div 4 \text{ A}$$

$$R = 3 \Omega$$

Pool links: T5D03 · T5D06

Worked pattern: find current

1 · FIND THE GIVEN

Given: 120 volts and 80 ohms

2 · NAME THE MISSING UNIT

Missing: amperes → current

3 · CHOOSE AND SOLVE

$$I = E \div R$$

$$I = 120 \text{ V} \div 80 \, \Omega$$

$$I = 1.5 \text{ A}$$

Pool links: T5D01 · T5D07

Worked pattern: find voltage

1 · FIND THE GIVEN

Given: 2 amperes and 10 ohms

2 · NAME THE MISSING UNIT

Missing: volts → voltage

3 · CHOOSE AND SOLVE

$$E = I \times R$$

$$E = 2 \text{ A} \times 10 \text{ } \Omega$$

$$E = 20 \text{ V}$$

Pool links: T5D02 · T5D12

Series: one path, same current

If the charge has only one route, every component in that route carries the same current.



Recognition rule

ONE PATH → SAME CURRENT through every component.

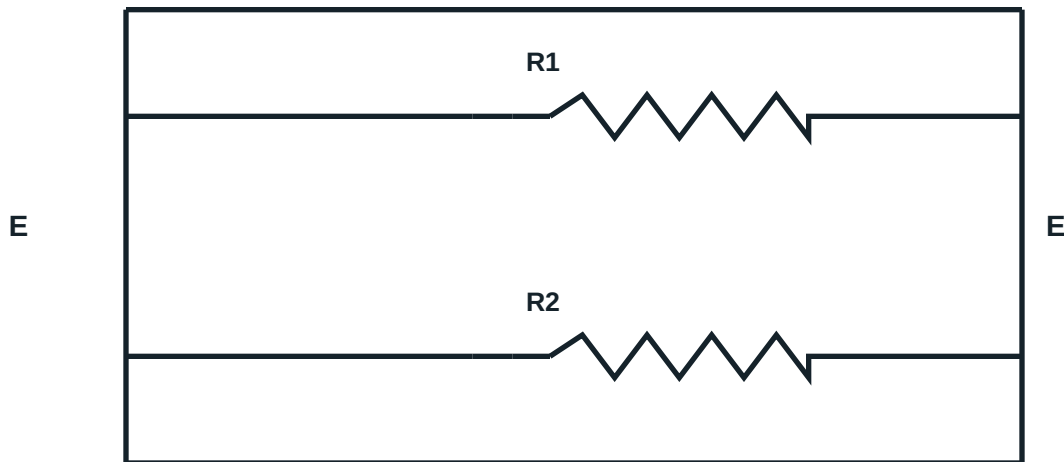
Do not overlearn

The prototype is testing recognition, not asking you to calculate equivalent resistance beyond the current pool's needs.

Pool link: T5D13

Parallel: shared endpoints, same voltage

If branches connect to the same two points, each branch has the same voltage across it.



Recognition rule

SAME TWO ENDPOINTS → SAME VOLTAGE across every branch.

Fast contrast

Series shares current. Parallel shares voltage.

Do not confuse path with position

SERIES: trace one uninterrupted route

One route through R1, then R2. Current cannot split. Question cue: “connected in series.”

PARALLEL: mark the two shared nodes

Each branch begins and ends at the same pair of connection points. Voltage across the branches matches.

The visual check

Cover the labels. Can you trace one path, or can you choose between branches? That matters more than whether the drawing is horizontal or vertical.

Meters follow the quantity

A meter must be connected in a way that lets it observe the quantity you are measuring.

VOLTMETER

Measures difference between two points.
Connect it across the component—in parallel.

AMMETER

Measures flow through a path. Put it into the path—in series.

Memory picture

Voltage looks ACROSS. Current goes THROUGH.

Pool links: T7D02–T7D03

Sixty-second self-check

1. A 12-volt source drives 4 amperes. What is the resistance?

2. A 10-ohm resistor carries 2 amperes. What voltage is across it?

3. In a series circuit, which quantity is the same through every component?

4. In parallel branches, which quantity is the same across each branch?

5. Does a voltmeter go in series or in parallel?

Answers and misconception check →

Answers: diagnose the mistake

1 • $3\ \Omega$

$R = E \div I = 12 \div 4$. If you multiplied, you answered a voltage question.

2 • $20\ \text{V}$

$E = I \times R = 2 \times 10$. The requested unit—volts—selects the formula.

3 • Current

Series has one path. The current does not split.

4 • Voltage

Parallel branches share the same two endpoints.

5 • Parallel

A voltmeter compares two points; an ammeter is inserted into the current path.

How the crosswalk behaves

The finished book teaches concepts in learning order. The crosswalk restores official traceability without turning the main text into 409 sequential mini-essays.

T5D01–T5D03	Formula chooser	pp. 3–4
T5D04–T5D12	Worked Ohm’s-law patterns	pp. 5–7
T5D13	Series recognition	p. 8
T5D14	Parallel recognition	pp. 9–10
T7D02–T7D03	Meter placement crossover	p. 11

Source chain

NCVEC 2026–2030 pool → technical source check → original explanation/diagram → claim ledger → ID crosswalk

Prototype source basis: NCVEC T5D and T7D; NIST Ohm’s-law relationship; OpenStax series/parallel circuit principles. Exact source records belong in the production claim ledger.